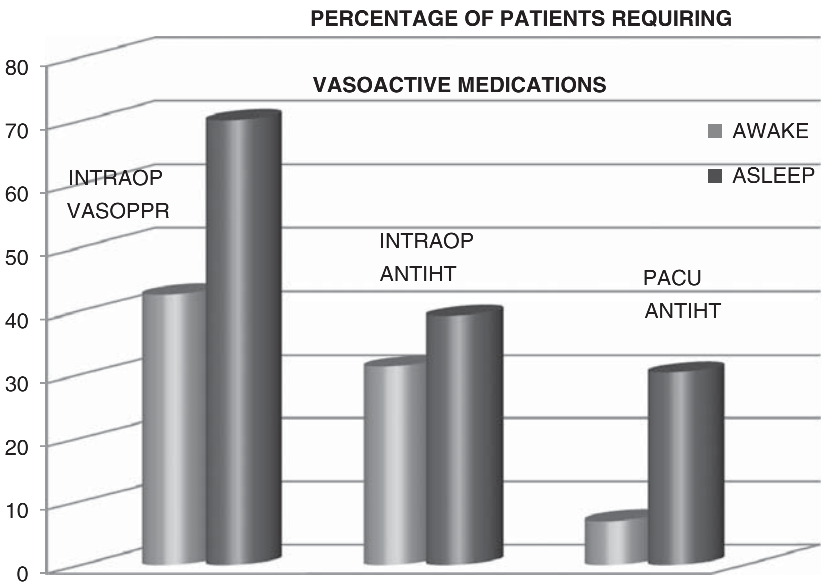

A101 Clopidogrel Resistance in Patients Requiring Endovascular Neurological Intervention

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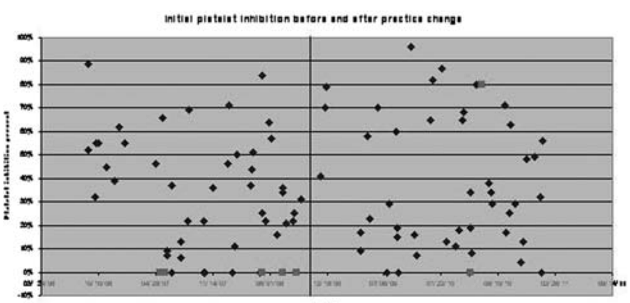
Introduction: Clopidogrel resistance, with a reported incidence of 52%, is increasingly important in vascular intervention because it may put patients at risk for perioperative thrombosis. We investigated the incidence of postoperative thrombosis in patients requiring endovascular carotid stent placement.

Methods: After IRB approval we retrospectively reviewed the charts of 96 patients requiring endovascular carotid stenting between 2006 and 2010. Each patient was treated with aspirin 325 milligrams (mg) daily and clopidogrel 75 mg daily for 5 days before intervention. On the day of the procedure, patients' blood was tested with a VerifyNow-P2Y12 clopidogrel assay (Accumetrics, San Diego, CA) to determine the effectiveness of clopidogrel. Patients were treated with an extra dose of clopidogrel 75 mg if they were found to be resistant (defined as $<40\%$ platelet inhibition) on the day of their surgery and the procedure commenced. After a number of thrombotic complications, practice changed in 2008 to try to increase the efficacy of clopidogrel in resistant patients. Therefore after August 2008, clopidogrel resistant patients were treated with increasing doses of clopidogrel daily until their assay showed $>20\%$ inhibition. In some patients, the assay never reached the 20% threshold, but only after the attempt to increase their percent platelet inhibition was their procedure performed. For purposes of data analysis, clopidogrel resistance is defined as initial percent platelet inhibition of $<20\%$.



Results: The main outcome variable was periprocedural thrombosis, which occurred in 7 patients. Statistically, it was insignificant whether the patient was treated before the practice change in 2008 or not. Prior to developing a logistic regression model, univariate analyses revealed that the following factors were associated with the outcome variable periprocedural thrombosis: use of statins ($P = 0.003$), clopidogrel resistance ($P = 0.004$), high diastolic blood pressure ($P = 0.04$), and history of stroke ($P = 0.09$). The logistic regression model ($P = 0.002$) identified the following variables as predictive of periprocedural thrombosis: clopidogrel resistance, lack of statin use, and higher diastolic blood pressure.

Conclusion: We show that clopidogrel resistance is associated with periprocedural thrombosis in patients after endovascular carotid intervention. Figure 1 graphically depicts all 96 patients as blue diamonds or pink squares, indicating their initial percent platelet inhibition by clopidogrel. The 7 pink squares depict those patients who experienced periprocedural thrombosis. The vertical line in the middle indicates the time of the practice change in 2008.

A102 The Application of Electro-Acupuncture Combined With Sevoflurane Anesthesia in Neurosurgery

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Object: To observe the effects of electro-acupuncture combined with sevoflurane anesthesia used in neurosurgery patients and the speed of recovery after surgery.

Methods: Eighty cases of supratentorial tumor resection were anesthetized with sevoflurane and randomly allocated into no treatment group (Group A), electro-acupuncture group (Group B). Han's acupoint nerve stimulator with 2/100 Hz frequency was used to stimulate the points. The patients in Group B received electro-acupuncture at FengChi (GB20) tou TianZhu (BL10) and CuanZhu (BL2) tou YuYao (EX-HN4) at the same side of the craniotomy before anesthesia induction. The stimulation is continued until the end of operation. All patients were induced with propofol 2 mg/kg, sufentanyl 0.3 μg/kg and vecuronium 0.1 mg/kg and maintained with sevoflurane. The end-tidal sevoflurane concentration, minimum alveolar concentration (MAC), bispectral index (BIS), changes of blood pressure and heart rate, and the speed of recovery were investigated in the 2 groups.

TABLE 2. End-tidal Sevoflurance Concentration Required to Maintain Stable Hemodynamics at Dierent Stages of Surgery

	Group A (n=40) (Acupuncture - Sevoflurane group)	Group B (n=40) (Sevoflurane group)	P	Reduction percentage (%)
Skin incision	1.60 ± 0.37	1.47 ± 0.33	0.214	8.12
Drilling skull	1.72 ± 0.43	1.56 ± 0.34	0.196	9.30
After skull drilling	1.74 ± 0.39	1.57 ± 0.36*	**0.041**	9.77
Opening the skull	1.75 ± 0.39	1.60 ± 0.40	0.056	8.57
Dura incision	1.79 ± 0.43	1.60 ± 0.36*	**0.022**	10.61
Intracranial period (10min)	1.78 ± 0.45	1.61 ± 0.32	0.057	9.55
Intracranial period (20min)	1.76 ± 0.44	1.61 ± 0.31*	**0.046**	8.52
Intracranial period (30min)	1.80 ± 0.44	1.67 ± 0.34*	**0.037**	7.22
Suturing dura	1.83 ± 0.39	1.69 ± 0.28	0.102	7.65
Sutuing skin	1.73 ± 0.37	1.66 ± 0.22	0.682	4.04

Data are mean ± SD or numbers. * $P < 0.05$ compared to the sevoflurane group.

Results: Hemodynamic variables were similar throughout surgery in both groups. The end-tidal concentration and MAC of sevoflurane were lower in Group B than in Group A, $P < 0.05$. The requirement of sevoflurane was reduced by an average of 8.34% in Group B. The BIS value was kept within 40̃50 in both groups, so as to insure a relatively anesthesia depth. The times to recovery of spontaneous respiration, extubation, opening eyes, voluntary movement and discharge from the recovery room were significantly shorter in Group B than in Group A, $P < 0.01$. Dysphoria, nausea and vomiting occurred in fewer patients in Group B.
Conclusion: Electro-acupuncture combined with sevoflurane used in neurosurgery can decrease the expired concentration and MAC of sevoflurane. It also speeds the recovery of anesthesia and improves the postoperative recovery of neurosurgery patients.
Keywords: Electro-acupuncture; Sevoflurane; Neurosurgery; Recovery.

A103 Combination of Standard Monitoring Parameters as a Reliable Measure of the Hypnotic Component of Anesthesia

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Introduction: Standard monitoring of anesthesia is mainly based on cardiovascular, respiration and to a certain extent motor response parameters, which provide information for assessing the hypnotic component of anesthesia. Specific parameter behavior and inter-patient variability requires a high level of the anesthesiologist experience for its evaluation. The present investigation evaluates ability and efficiency of the BIS and an indicator that combines different standard parameters using an adaptive fuzzy inference to indicate "depth of anesthesia."
Methods: After ethics committee approval and written informed consent, 263 adult patients undergoing surgery under general anesthesia were included in a study conducted in 6 European centers.[1] Patients were randomly assigned to 1 of 11 anesthetic groups, consisting of different analgesic and hypnotic drugs for induction/maintenance phase. Standard parameters and EEG were continuously recorded and stored together with patient data (eg, age, weight, ASA, drug protocol). During anesthesia induction, Tunstall isolated forearm technique was performed to detect loss of consciousness (LOC). After skin incision, the anesthetic concentration was increased until EEG burst suppression (BS).

At the end of surgery, drugs were discontinued until return of consciousness (ROC). An adaptive fuzzy inference based on a Takagi Sugeno Kang model (TSK)[2] was used for data driven combination of standard parameters (heart rate, blood pressure, inspiratory and expiratory gas concentrations, pulmonary peak pressure), patient data and drug protocol (threefold cross validation). The adaptive algorithm is able to provide indicator values even in case of partially available input data (eg, caused by measurement failures, equipment problems and human errors[3]). TSK and BIS (calculated offline[4]) were analyzed during consciousness (before LOC, after ROC), unconsciousness (after LOC, before ROC), general anesthesia and BS. Prediction probability (PK) including 95% bootstrap confidence intervals indicates the indicators ability to separate anesthetic levels.
Results: The TSK indicator leads to a PK of 0.90 (0.88 to 0.91) for separation of consciousness, unconsciousness, general anesthesia and BS. This is significantly higher than BIS with PK of 0.80 (0.78 to 0.81). Thereby, the TSK provides values even in case of false/missing standard parameter values (14.3% occurrence).
Conclusions: Single standard parameters provide partial information about the hypnotic component and have to be interrelated to assess "anesthetic depth." The presented approach allows combining this individual information by a data driven adaptive fuzzy inference, so that the resulting indicator includes anesthesiologist's expertise about parameter relevance. Results show that separation of consciousness, unconsciousness, general anesthesia and BS can be sufficiently revealed by the obtained indicator even though individual parameters are partially acquired. In particular, based on data of a multicenter study the indicator leads to a better separation than the electroencephalogram based BIS.

References:
1. *Anesthesiology*. 2006;105:A1553.
2. *IEEE Trans Syst Man Cybern*. 1993;23.
3. *BJA*. 2002;89(6):825-831.
4. *Anesth Analg*. 2007.

A104 Intracerebral Hemorrhage—Does eGFR Affect Hospital Mortality?

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Background: Intracranial hemorrhage (ICH) is the second most common cause of stroke, with an incidence of 24.6 per 100,000 patient-years. The median case fatality is 40% at 1 month, with 12% to 39% of survivors independent of care after the stroke. Estimated glomerular filtration rate (eGFR) is a measure of renal function with an eGFR $< 60\,\mathrm{mL/min/1.73\,m^2}$ diagnostic of chronic kidney disease. eGFR is associated with worse outcome after cardiac surgery and acute stroke. Our aim was to determine if an association between eGFR and outcome after ICH exists.
Methods: This is a retrospective review of data for all patients admitted to our ITU with an ICH from 1.11.09 to 31.1.11. Data collected included: patient demographics, admission severity of illness score (APACHE II and SOFA), duration of mechanical ventilation and ITU stay, eGFR on admission. Survival (dead or alive) was determined at discharge from hospital. eGFR was calculated using the 4 variable Modification of Diet in Renal Disease formula. Sub-group analysis compared patients with: eGFR $< 60\,\mathrm{mL/min/1.73\,m^2}$, 60 to 89 $\mathrm{mL/min/1.73\,m^2}$ and eGFR $\geq 90\,\mathrm{mL/min/1.73\,m^2}$.
Results: Sixty-nine patients were included. Mean age was 52.1 years (SD ± 14.6). There was no difference in demographics or admission APACHE II/SOFA score between the groups. Overall patient mortality was 31.9%. Total 35.3% of patients with eGFR $< 90\,\mathrm{mL/min/1.73\,m^2}$ died compared with 28.5% with eGFR $\geq 90\,\mathrm{ml/min/1.73\,m^2}$ ($P = 0.3667$ for difference). Subgroup analysis also showed no difference in mortality (Table 1).

No statistically significant difference in median days ventilated [5 (0 to 29) vs. 3.5 (0 to 28) d, $P = 0.3584$], or median duration of ITU stay [5 (1 to 31) vs. 7 (1 to 29) d $P = 0.2111$] between the patients with eGFR $\geq 90\,\mathrm{mL/min/1.73\,m^2}$ and those with eGFR $< 90\,\mathrm{mL/min/1.73\,m^2}$. Subgroup analysis showed no statistically significant difference between the 3 groups in duration of ITU stay or mechanical ventilation.
Conclusions: We found no significant difference in hospital mortality, duration of mechanical ventilation or ITU stay between ICH patients admitted to ITU with an eGFR $\geq 90\,\mathrm{mL/min/1.73\,m^2}$ and eGFR

